An established traffic backbone for an elementary VSL task

Literature comparison and formed modeling backbone — revised draft, 21 October 2026

1. What a suitable backbone must provide

Can an established variable speed limit (VSL) model support the information, control and gradient questions with few traffic-model changes? Consider an ending lane E, its adjacent continuing lane A and farther lane B. Initial states can share the same E and combined A+B traffic at every road position while differing in A/B loading. We ask whether that difference changes merge service, useful common-speed advice and its sensitivity. A suitable backbone must represent the distinction, conserve traffic, respect receiving space and count every initial unit's time until exit. Simplicity concerns the traffic rules we must author and justify, not citation count.

2. What the closest published models offer

Aggregate VSL models support questions about traffic location or history; lane models additionally resolve A/B loading. The inspected starting points offer the following tradeoff.

Table 1. Published starting points and their fit to the question

Published model	What is already supplied	Direct finding for this review
Jin and Jin [1]	Spatial and link-average traffic; VSL entry control, capacity drop and upstream holding queues.	Simpler if the question concerns *where* traffic lies along the road. It changes the A/B information question.
Wei, Han and Wang [2]	Link traffic retaining entry/speed history, interacting speed cohorts and capacity drop.	A strong aggregate alternative, but no explicit A/B loading. Native advice acts on entering traffic.
Nagalur Subraveti et al. [3]	Successive lane-cell densities, lane-change incentives and a lane-drop case; Han congestion lineage [4].	Selected lane backbone. Adopt Han's VSL sending restriction [5] and explicitly replace movement realization.
Carey et al. [6]	Multilane cell transmission, mandatory movements and receiving-allocation alternatives.	Supports conservative service and forward fallback. More VSL/lane-end task closure would be needed as the main backbone.

Selection. Use the Subraveti lane-density framework and its Han congestion/VSL lineage, with one explicit conservative movement replacement. This retains the chosen lane-information question. Jin–Jin remains a genuine simpler route if the question changes to spatial information. The comparison concerns inspected methods, not a globally best-model claim; the exact cohort/control scenario and movement composition below are ours.

3. The formed modeling backbone

3.1. Physical picture, task and root assumptions. This is a *lane-resolved cell-transmission-type (CTM) model* using Han's extended first-order budgets. Physical cells retain density and finite storage throughout the road; there is no separate link transmission model (LTM) link. E can change into A at upstream cell boundaries; traffic in the final E cell must merge at the exit. Unserved mass stays in its current cell, so congestion can extend upstream. The causal chain is:

$$\boxed{\text{Lane stocks}} \longrightarrow \boxed{\text{Desires and budgets}} \longrightarrow \boxed{\text{Shared movements}} \longrightarrow \boxed{\text{New stocks}}$$

Use $N = 10$ sections, $\Delta x = 0.1\,\text{km}$ and $\Delta t = 1/600\,\text{h}$ (6 s). At step k, density $\rho_{i\ell}^k$ and continuous stock $n_{i\ell}^k = \Delta x \rho_{i\ell}^k$ describe section i, lane $\ell \in \{E, A, B\}$. Initialize a finite cohort independently of constant common advice $v \in [30, 60]\,\text{km/h}$, with no later arrivals. Free A/B outlets complete trips. Assume common lane flow–density parameters, fixed geometry, old traffic/space for every movement, equal positive priorities and no extra merging-headway multiplier. The step satisfies $u_0 \Delta t, w \Delta t \leq \Delta x$; u_0 is nominal free speed and w the backward-wave speed magnitude.

3.2. How much traffic can move or be received? The full Han budgets [4], [5] give a sending rate D and a shared receiving rate R. The discharge ceiling $Q(x)$ equals nominal capacity C below critical density ρ_c, then falls linearly to $(1-\delta)C$ at jam density k_J:

$$C = \frac{u_0 w k_J}{u_0 + w}, \quad \rho_c = C/u_0, \quad Q(x) = \min\left\{C, C\left[1 - \delta\frac{x-\rho_c}{k_J-\rho_c}\right]\right\}, \tag{1}$$

$$D_{i\ell} = \min\{v\rho_{i\ell}, Q(\rho_{i-1,\ell})\}, \quad R(x,y) = \min\{Q(x), w(k_J-y), w(k_J-x)+w_d(x-y)\}. \tag{2}$$

Here x, y are the current upstream/downstream densities in the receiving lane. The three receiving bounds express discharge capacity, downstream space and Han's additional congestion-wave constraint. Its fixed recovery slope is $w_d = Q_J/(k_J - Q_J/u_0)$, $Q_J = (1-\delta)C$. The member uses $u_0 = 60$, $w = 20\,\text{km/h}$, $k_J = 150\,\text{veh/km/lane}$ and $\delta = 0.1$; initial densities lie in $[0, k_J]$. Available amounts are $d_{i\ell} = \Delta t D_{i\ell}$ and $r_{ij} = \Delta t R(\rho_{ij}, \rho_{i+1,j})$: all straight and diagonal arrivals to target j share that *one* r_{ij}. At the outlets use $r_{Nj} = \Delta t R(\rho_{Nj}, 0)$, $j = A, B$; upstream boundary densities are zero with no inflow. Advice caps sending; receiving changes through the evolving densities, not directly through v.

3.3. Why and where does early merging occur? The retained Subraveti incentive framework [3] compares perceived lane densities K, using current and two downstream cells with weights 2:2:1 [7]. For a nonempty origin and adjacent destination,

$$P_{i,\ell j} = \left[\frac{I_{i,\ell j}K_{i\ell} - K_{ij}}{K_{i\ell} + K_{ij}}\right]_+, \qquad I = 1 + I^r + I^{kr} + I^{coop}, \qquad [x]_+ = \max(x, 0). \tag{3}$$

Less crowded destinations attract traffic. Within the last $D_r = 0.75\,\text{km}$, E-to-A route bias grows as $I^r = (1 - g_i/D_r)^3$, where $g_i = (N-i)\Delta x$; it is zero farther upstream. This adds urgency, not a fixed merge fraction or a hard start line: density incentives can cause earlier E-to-A movement. Keep-right and cooperation use the member's declared mode/direction conventions; free A can be encouraged toward B to make room. Missing downstream E cells are omitted and weights renormalized; A/B lookahead extends with zero outlet density. Empty origins make no requests. E, A and B name current lanes, so early-merged traffic becomes A.

3.4. Which attempts succeed, and what happens at the choke? Our replacement of native same-cell lateral/adjusted-forward coupling separates desire from service. Requests use existing forward or adjacent cells/outlets. At one boundary, temporarily omit i. Normalize lateral desires, leave the rest for straight travel, and form requests m from each donor's one sending amount:

$$b_{\ell j} = \frac{P_{\ell j}}{\max(1, \sum_{h\neq\ell} P_{\ell h})} \ (j \neq \ell), \quad b_{\ell\ell} = 1 - \sum_{j\neq\ell} b_{\ell j}, \quad m_{\ell j} = d_\ell b_{\ell j}. \tag{4}$$

Adapt Carey receiving/fallback principles and Wright priority sharing [6], [8]: each target serves requests at a common capped level,

$$z_{\ell j} = \min(m_{\ell j}, \lambda_j), \qquad \sum_\ell z_{\ell j} = \min\left(r_j, \sum_\ell m_{\ell j}\right). \tag{5}$$

Requests below the shared level are fully served; unused entitlement returns to the others. Residual donor traffic gets one straight fallback y_ℓ, the minimum of its unused sending and its own lane's unused receiving amount. Thus $f_{\ell j} = z_{\ell j} + \mathbf{1}_{\{j=\ell\}} y_\ell$ is realized movement. An unsuccessful E-to-A attempt can therefore continue in E if space remains. At final E override $b_{EA} = 1$ and $y_E = 0$: E must use A's outlet or wait. It competes with A and any B-to-A request for the same outlet budget. Equal priorities are not a constant realized merge proportion; demand and receiving state determine the flow. E's use leaves less service for the other requests, whose retained stock can restrict later receiving. There is no separate speed-mismatch or acceleration-recovery meter.

Every movement advances one section, diagonally when changing lanes. Apply all flows from the old state simultaneously:

$$n_{i\ell}^{k+1} = n_{i\ell}^k - \sum_j f_{i,\ell j}^k + \sum_h f_{i-1,h\ell}^k. \tag{6}$$

Set $f_0 = 0$; outlet flows f_N complete trips. Matched additions/subtractions conserve mass, while donor/receiver limits protect stock and storage.

3.5. Complete objective, information comparison and sensitivity. Minimize total system time J, counting all initial mass until exit. Once remaining mass is below $\Delta x \rho_c$, all requests receive service and a unit in section i has remaining mean time $(N - i + 1)\Delta x/v$. Choose a fixed index K_* inside that regime. With $h_{i\ell} = N - i + 1$ and $\mathbf{1}^\top n^k$ the total unfinished mass, the exact prefix-plus-residual objective is

$$J(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top n^k + \frac{\Delta x}{v} h^\top n^{K_*}. \tag{7}$$

The residual counts future time rather than deleting small remaining stock. Write the complete update as $n^{k+1} = F(n^k, v)$. Between branch switches, $s^0 = 0$, $s^{k+1} = F_n^{[k]} s^k + F_v^{[k]}$ propagates $s^k = \partial n^k / \partial v$ through budgets, desires, allocation and stock. Then

$$J'(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top s^k + \frac{\Delta x}{v} h^\top s^{K_*} - \frac{\Delta x}{v^2} h^\top n^{K_*}. \tag{8}$$

These terms measure earlier traffic changes, residual-stock changes and direct residual travel time. Hold a valid K_* fixed locally; at native mode or active-constraint switches a two-sided derivative can fail.

The information comparison uses this same physical model and objective: one observer sees E and A+B at each section, another sees each lane. Admit model-aware inference from allowed histories; choose advice before the subsequent revealing discharge. A compact observer adds the final-A density, sufficient on the existing two-state diagnostic. Separately, compare gradients with bounded scalar search using the same information, and retain maximum advice as a reference. These definitions isolate what information is needed from how the advice is optimized. The companion formulation specifies all bias branches and finite-domain conventions.

References

[1] H.-Y. Jin and W.-L. Jin, "Control of a lane-drop bottleneck through variable speed limits," arXiv:1310.2658, 2013 (archived manuscript dated Oct. 29, 2018). [Online]. Available: `https://arxiv.org/abs/1310.2658`.

[2] L. Wei, Y. Han, and M. Wang, "A first-order link-based flow model with variable speed limits and capacity drops for freeway networks," *Comput.-Aided Civ. Infrastruct. Eng.*, vol. 40, no. 21, pp. 3200–3217, 2025, doi: 10.1111/mice.13492.

[3] H. H. S. Nagalur Subraveti, V. L. Knoop, and B. van Arem, "First order multi-lane traffic flow model—an incentive based macroscopic model to represent lane change dynamics," *Transportmetrica B, Transp. Dyn.*, vol. 7, no. 1, pp. 1758–1779, 2019 (archived article header: 2020), doi: 10.1080/21680566.2019.1700846.

[4] Y. Han, Y. Yuan, A. Hegyi, and S. P. Hoogendoorn, "New extended discrete first-order model to reproduce propagation of jam waves," *Transp. Res. Rec.*, vol. 2560, pp. 108–118, 2016, doi: 10.3141/2560-12.

[5] Y. Han, A. Hegyi, Y. Yuan, S. Hoogendoorn, M. Papageorgiou, and C. Roncoli, "Resolving freeway jam waves by discrete first-order model-based predictive control of variable speed limits," *Transp. Res. C, Emerg. Technol.*, vol. 77, pp. 405–420, 2017, doi: 10.1016/j.trc.2017.02.009.

[6] M. Carey, C. Balijepalli, and D. Watling, "Extending the cell transmission model to multiple lanes and lane-changing," *Netw. Spat. Econ.*, vol. 15, pp. 507–535, 2015, doi: 10.1007/s11067-013-9193-7.

[7] H. H. S. Nagalur Subraveti, *Lane-specific Traffic Flow Control*, Ph.D. dissertation, Delft Univ. Technol., Delft, The Netherlands, 2021, ch. 4, doi: 10.4233/uuid:e405a549-b1fe-4fa1-8090-4429612359d9.

[8] M. A. Wright, G. Gomes, R. Horowitz, and A. A. Kurzhanskiy, "On node models for high-dimensional road networks," *Transp. Res. B, Methodol.*, vol. 105, pp. 212–234, 2017, doi: 10.1016/j.trb.2017.09.001.